MIL-R-16417J
31 October 1986
SUPERSEDING
MIL-R-16417H
15 October 1975
(See 6.6)

MILITARY SPECIFICATION

REPAIR KITS, CO_2, INFLATABLE LIFE PRESERVERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for repair kits for repairing carbon dioxide (CO_2) inflatable life preservers.

1.2 Classification. The repair kits shall be of the following types, as specified (see 6.2):

Type I - Patches and patching material (orange).
Type II - Patches and patching material (gray).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

GGG-K-275 - Key, Socket Head Screw and Key Sets Socket Head (Wrench Hexagon, Spline and Square).
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-A-5540 - Adhesive, Polychloroprene.
MIL-C-17564 - Cloth, Holland.
MIL-C-19002 - Cloth, Coated, and Strip, Coated Cloth-Polychloroprene on Nylon.
MIL-L-24247 - Life Preserver, Vest Type, U.S. Navy, Mark I.
MIL-S-45179 - Stitchers, Cementing and Vulcanizing, and Rollers, Hand.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials. The materials shall conform to the requirements specified herein or as specified (see 6.2).

3.3 Container and parts. Repair kits shall consist of a water-resistant fiber container with a screw cap, complete with the following parts contained therein:

(a) One hundred patches, 1.5 by 2 inches.
(b) One piece of patching material, 10 by 24 inches.
(c) One piece of holland cloth, 10 by 24 inches.
(d) One steel roller conforming to type I, size 1 in accordance with MIL-S-45179.
(e) Three relief valve diaphragms in accordance with MIL-L-24247.
(f) Twenty-four upper gaskets for CO_2 system in accordance with MIL-L-24247.
(g) Twenty-four lower gaskets for CO_2 system in accordance with MIL-L-24247.

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(h) Twelve hex caps for CO_2 system in accordance with MIL-L-24247.
(i) Four one-way check valves for manifold of CO_2 system in accordance with MIL-L-24247.
(j) Four yards of shear wire in accordance with MIL-L-24247.
(k) One valve repair tool.
(l) Two relief valve cap removal tools.
(m) Five socket-head screw keys, size 0.050.
(n) The container closure shall be sealed with a pressure-sensitive water-resistant tape and the container shall show no evidence of leakage when subjected to the test specified in 4.4.3.
(o) Application instruction sheet.

3.4 Patches and patching material. Patches and patching material shall be made of coated cloth in accordance with type I of MIL-C-19002, except for color. Color of patches and patching material for types I and II shall conform to color numbers 12246 and 36081, respectively, of FED-STD-595.

3.5 Holland cloth. Holland cloth shall be in accordance with MIL-C-17564.

3.6 Relief valve cap removal tool. The relief valve cap removal tool shall be PAM Co., Inc., part number PRV-200SW, or equal.

3.7 Valve repair tool and socket head screw key. The valve repair tool shall be a Schrader part 3522, or equal. The screw key shall conform to type I, class 1, size 0.050 inch of GGG-K-275.

3.8 Instructions. An application instruction sheet with the following details shall be inserted in each container:

3.8.1 Instructions for repairing small damaged areas (up to and including 0.25 inch):

WARNING: DO NOT USE SOLVENTS NEAR OPEN FLAME, HEAT, OR SPARKS. AVOID PROLONGED CONTACT WITH SKIN OR BREATHING OF FUMES. USE ONLY IN WELL VENTILATED AREAS.

(a) Wipe the repair area and coated side of patch with methyl chloroform and allow to dry thoroughly. Patch shall be applied in such a manner as to provide not less than 0.625-inch lap on all sides of the damaged area.
(b) MIL-A-5540, class 3, form B, adhesive shall be applied in four coats (brushing thoroughly) to repair area and three coats of adhesive to coated side of patch. Allow each coat to dry 10 minutes.
(c) Apply coated side of patch to repair area and roll down thoroughly with roller. Allow to dry approximately 15 to 30 minutes before inflating preserver.

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3.8.1.1 Instructions for repairing large damaged areas (over 0.25 inch):

WARNING: DO NOT USE SOLVENTS NEAR OPEN FLAME, HEAT, OR SPARKS. AVOID PROLONGED CONTACT WITH SKIN OR BREATHING OF FUMES. USE ONLY IN WELL VENTILATED AREAS.

(a) Damaged areas over 0.25 inch and less than 4 inches shall be lengthened to allow inner patch to be inserted.
(b) Cut a piece of patch material at least 1 inch larger all around than the damaged area.
(c) Wipe the coated side of the patch, and also an area inside the damaged area large enough to take a patch, with methyl chloroform and allow to dry thoroughly.
(d) MIL-A-5540, class 3, form B, adhesive shall be applied in three coats to one half of the inside surface of the repair area, parallel to the length of the repair.
(e) Apply three coats of adhesive to the coated half of the patch. Allow each coat to dry 10 minutes.
(f) Place a piece of holland cloth on the adhesive coated side of patch (the holland cloth should be slightly wider than the adhesive coated portion) so that about 2 inches of adhesive are exposed at one end of the patch.
(g) Fold patch just sufficiently to insert through damaged area and line up center of patch with center of repair area.
(h) Press the exposed adhesive coated portion of the patch to the matching portion of the damaged area. Slowly pull the holland cloth from the patch and at the same time press the exposed adhesive coated area of the patch to the side of the damaged area so that it covers the damaged area evenly and without wrinkles.
(i) MIL-A-5540, class 3, form B, adhesive shall be applied in three coats to the other half of the patch. Allow each coat to dry 10 minutes. Separate patch from repair area by placing a piece of holland cloth on the adhesive coated area.
(j) MIL-A-5540, class 3, form B, adhesive shall be applied in three coats to the matching portion of repair area. Allow each coat to dry 10 minutes.
(k) Slowly pull the holland cloth from the patch and press the patch to the repair area. Roll down thoroughly with roller.
(l) Apply an outside patch following the procedures specified in the instruction for repairing small damaged areas. Allow to dry approximately 15 to 30 minutes before inflating preserver.

3.8.1.2 Instructions for repairing inflation system:

(a) To change the one-way check valve in the CO_2 manifold, the CO_2 cylinder holder shall first be removed.
(b) Using the valve repair tool, turn out the one-way check valve and replace with a new valve.
(c) Before replacing the CO_2 cylinder holder, remove and discard the lower gasket on the manifold and replace it with a new washer of the same type.

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(d) Replace CO_2 cylinder holder on manifold; place new upper gasket on manifold. Inspect hex cap and if defective, replace with new cap.
(e) Replace CO_2 cylinders into holder and safety wire CO_2 lever in position with shear wire.

3.8.1.3 Instructions for repairing pressure relief valve:

(a) Remove two set screws in pressure relief valve cap, using the socket-head screw key.
(b) Using the relief valve-cap removal tool, turn the cap counter clockwise and remove.
(c) Remove spring, retaining ring and old diaphragm from valve body.
(d) Insert new diaphragm original retaining ring and original spring into valve body.
(e) Replace original valve cap, turning clockwise as far as it will go.
(f) Replace and tighten original set screws.

3.9 Workmanship. The finished product shall be free from defects affecting the appearance or which may impair serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. When required (see 6.2), inspection of the first article sample shall be made of a complete repair kit for all provisions of this specification applicable to end product examinations specified herein (see 3.1).

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4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Lot. For purposes of inspection, a lot shall consist of all complete repair kits of the same type offered for delivery at one time.

4.4.1.2 Sampling for visual and dimensional examination. A random sample of complete repair kits shall be selected from each lot at inspection level II of MIL-STD-105, and shall be subjected to the visual and dimensional examination of 4.4.2. The acceptable quality level (AQL) shall be 1.5.

4.4.1.3 Sampling for performance test. One container (see 3.3) shall be selected from each random sample for testing in accordance with 4.4.3.

4.4.2 Visual and dimensional examination. Each of the sample kits selected in accordance with 4.4.1.2 shall be visually and dimensionally examined to verify conformance to this specification as to materials, workmanship, and completeness. Any kit in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective kits in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for examination provided the contractor has repaired or removed all nonconforming kits.

4.4.3 Performance (leak) test. Leak test shall be performed in accordance with MIL-P-116 utilizing the vacuum chamber technique, except that the submersion test may be used in lieu of the vacuum chamber technique. Rejection criteria shall be as specified in MIL-P-116 for the test technique applied.

4.5 Inspection of packaging. Sample packages and packs and the inspection of the preservation-packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 Packaging. (Not applicable (see 3.3)).

5.2 Packing. Packing shall be level B or C as specified (see 6.2).

5.2.1 Level B. Repair kits shall be packed in quantities of 36 kits per container in fiberboard boxes conforming to class weather-resistant, grade V, of PPP-B-636, or class 2 of PPP-B-640. Box variety and style shall be at the contractor's option. Boxes shall be closed, waterproofed, and reinforced with tape as specified in the appendix to the applicable box specification with method V closure applying to PPP-B-636 containers.

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5.2.2 Level C. Repair kits shall be packed in quantities of 36 kits per container in fiberboard boxes conforming to class domestic of PPP-B-636, or class 1 of PPP-B-640. Box grade, variety, and style shall be at the contractor's option. Boxes shall be closed, strapped, or banded in accordance with the appendix to the applicable box specification with method I closure applying to PPP-B-636 containers.

5.3 Palletized and containerized unit loads. When specified (see 6.2), kits packed as specified in 5.2 shall be palletized or containerized in accordance with MIL-STD-147.

5.4 Marking. In addition to the marking required in the contract or order (see 6.2), exterior containers, palletized and containerized loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The repair kits covered by this specification are intended primarily for repairing CO_2 inflatable life preservers.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) If first article inspection is required (see 3.1 and 4.3).
(d) Materials, if other than specified (see 3.2).
(e) Level of packing required (see 5.2).
(f) When palletized loads are required (see 5.3).
(g) Marking requirements (see 5.4).

6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of the samples specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.5 Subject term (key word) listing.

Holland cloth
Patches
Repair kits

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Air Force - 99

Review activities:
DLA - CS
Air Force - 84

User activity:
Army - ME

Preparing activity:
Navy - SH
(Project 4220-0323)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-16417J	REPAIR KITS, CO_2, INFLATABLE LIFE PRESERVERS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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